



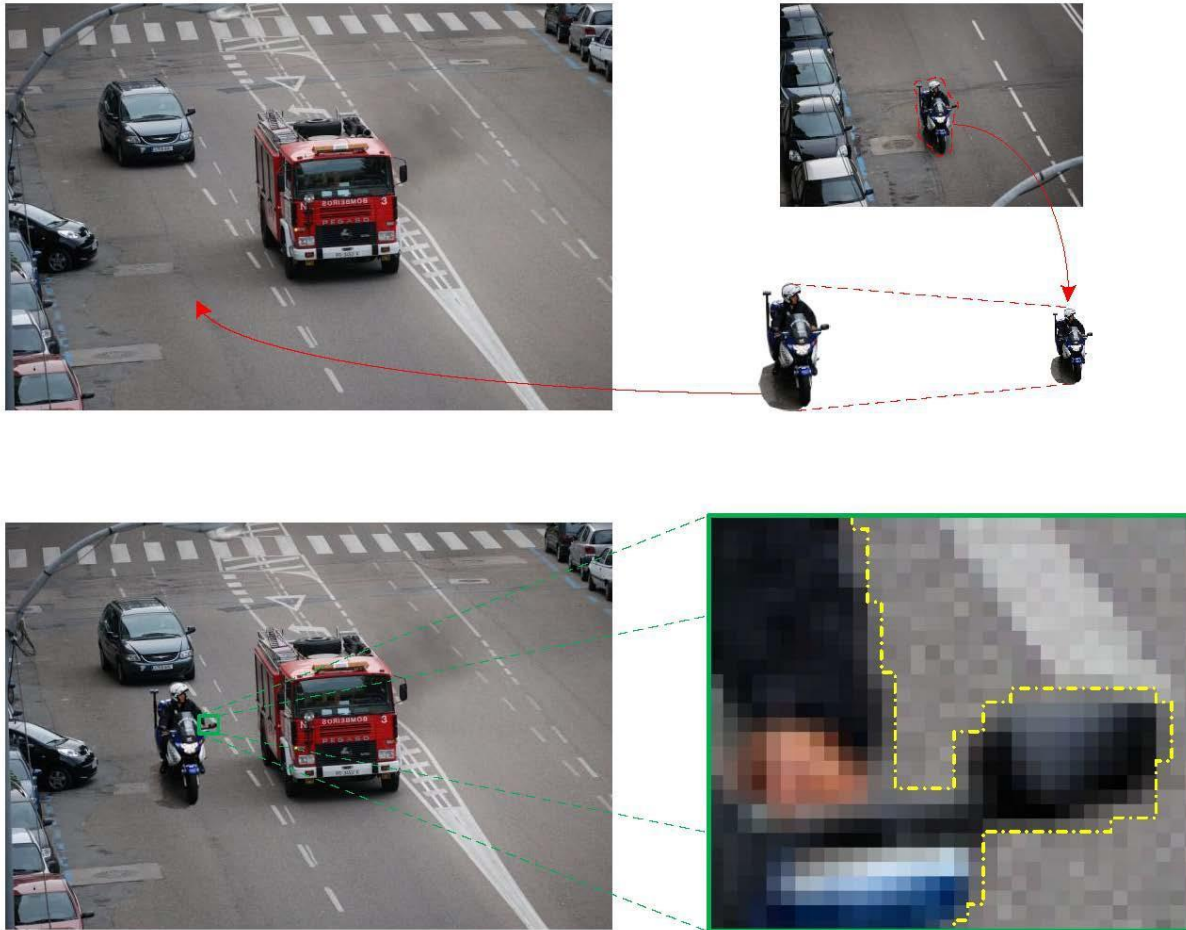
Image Editing Forensics

Taras Holotyak

Agenda

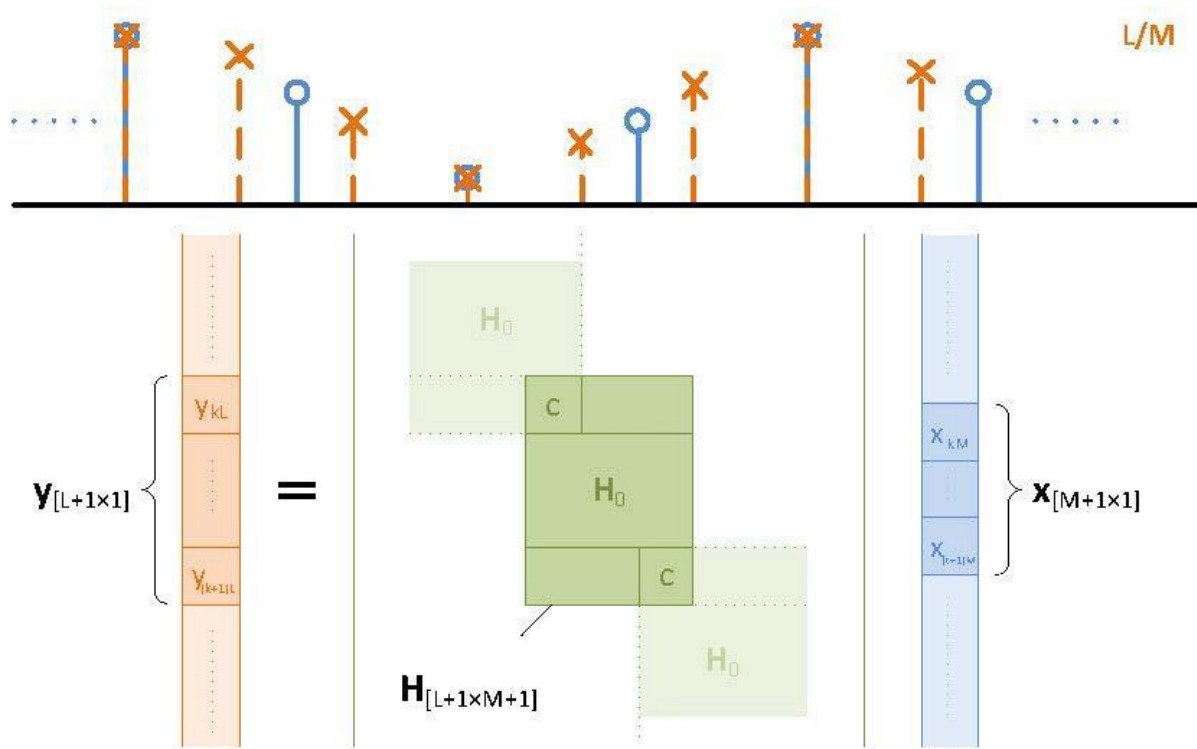
- Traces left by image editing
 - Resampling detection
 - Detection of contrast enhancement
 - ✓ global contrast enhancement
 - ✓ local contrast enhancement
 - Histogram equalization
- Tampering detection

Detecting traces of resampling



Detecting traces of resampling

Resampling by L/M



Detecting traces of resampling

Resampling factor estimation

- Writing the observed samples as a linear combination of the original samples

$$\mathbf{y} = \mathbf{H} \cdot \mathbf{x}$$

- The matrix for $L/M=4/3$ has the form

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & & \\ 0.25 & 0.75 & 0 & 0 & & \\ 0 & 0.50 & 0.50 & 0 & & \\ 0 & 0 & 0.75 & 0.25 & & \\ 0 & 0 & 0 & 1 & & \\ & & & & \ddots & \end{bmatrix}$$

- The matrix for $L/M=2$ has the form

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & & \\ 0.50 & 0.50 & 0 & & \\ 0 & 1 & 0 & & \\ 0 & 0.50 & 0.50 & & \\ 0 & 0 & 1 & & \\ & & & \ddots & \end{bmatrix}$$

Detecting traces of resampling

Resampling factor estimation

- If the length of $\mathbf{x}$ is N , then the length of $\mathbf{y}$ is approximately NL/M , so the size of $\mathbf{H}$ is approximately $(NL/M) \times N$. Then, for large N , there must be linearly dependent rows in $\mathbf{H}$
- Consider the matrix for $L/M = 4/3$. The second row $\mathbf{h}_2$ can be written as a linear combination of $\mathbf{h}_1$, $\mathbf{h}_3$, $\mathbf{h}_4$ and $\mathbf{h}_5$:

$$\mathbf{h}_2 = 0.25\mathbf{h}_1 + 1.5\mathbf{h}_3 - \mathbf{h}_4 + 0.25\mathbf{h}_5$$

- This implies that

$$y_2 = 0.25y_1 + 1.5y_3 - y_4 + 0.25y_5$$

- This relationship repeats periodically, i.e.,

$$y_{kL+2} = 0.25y_{kL+1} + 1.5y_{kL+3} - y_{kL+4} + 0.25y_{kL+5}$$

Detecting traces of resampling

Resampling factor estimation

- Assuming there is Gaussian noise in $\mathbf{y}$

$$f_{Y_i|\mathbf{b}_p}(y_i|\mathbf{b}_p) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{|y_i - \sum_{k=-P}^P y_{i-k} b_{p,k}|^2}{2\sigma^2}\right)$$

with $\mathbf{b}_p$ $p=i \bmod L$, the vector describing the linear combination

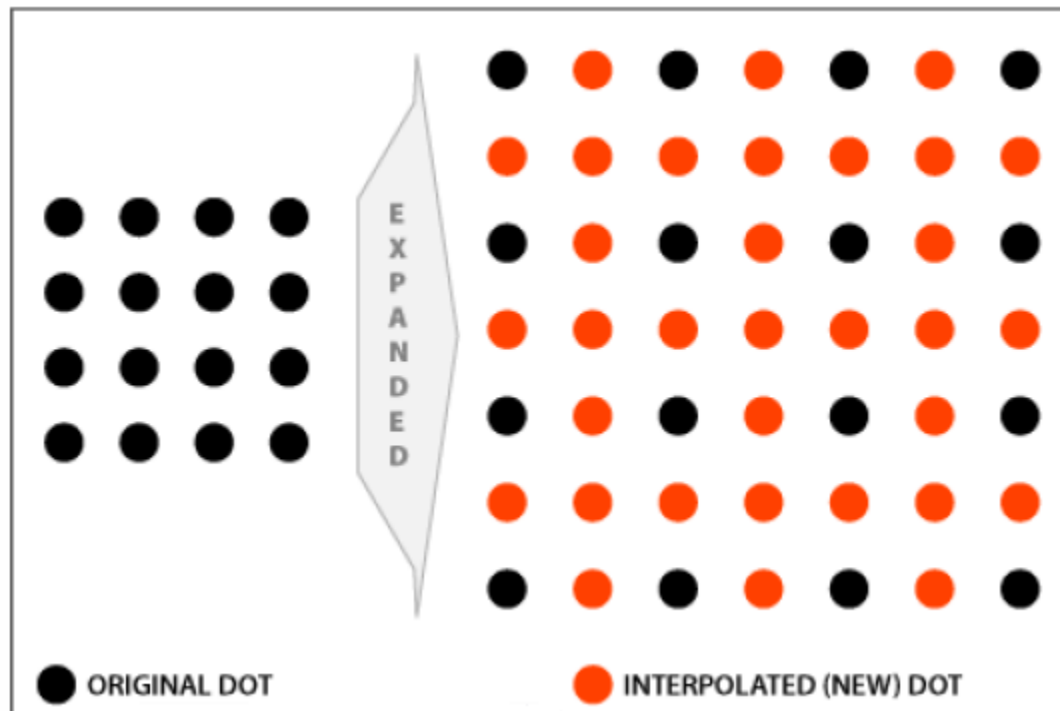
- In the case of no interpolation Y_i is assumed to be uniform over its domain
- Since neither $\mathbf{b}_p$ nor σ^2 are known, they can be estimated using the EM algorithm:
 - **E-step**: probability of Y_i given $\mathbf{b}_p$ and σ^2
 - **M-step**: estimate $\mathbf{b}_p$ as the minimizer of

$$\sum_i f_{Y_{iL+p}}(y_{iL+p}) \left(y_{iL+p} - \sum_{k=-P}^P y_{iL+p-k} b_{p,k} \right)^2$$

and σ^2 is the minimum cost

Detecting traces of resampling

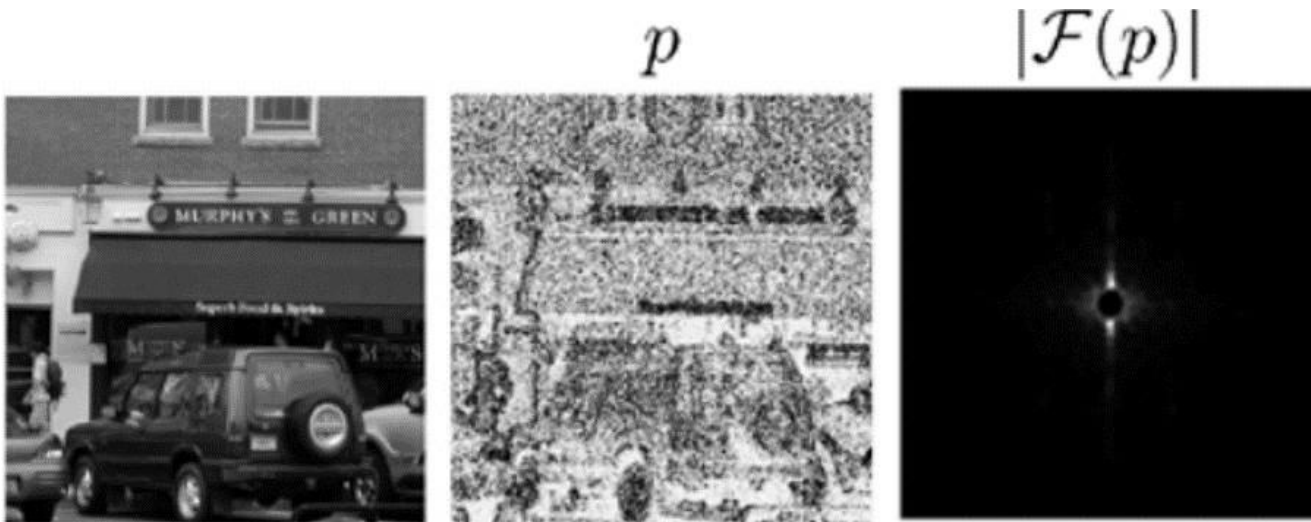
Resampling factor estimation



Detecting traces of resampling

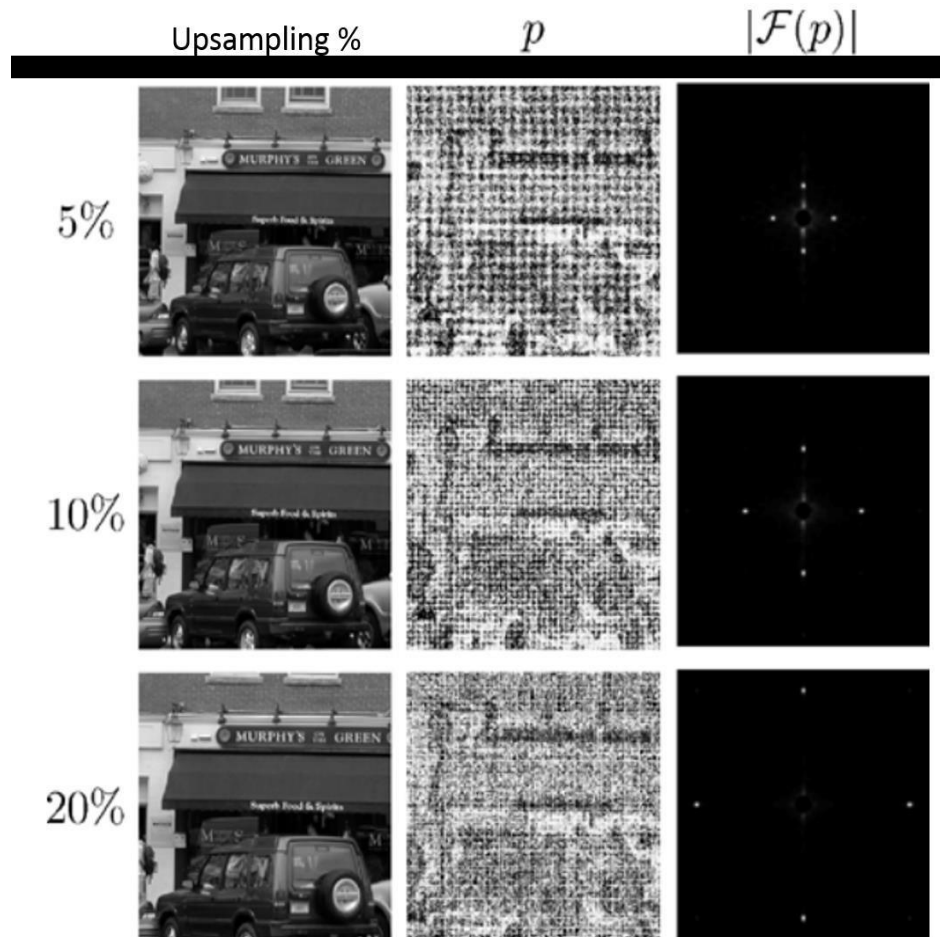
Resampling factor estimation

- The probabilities for each y_i can be represented in a probability map
- These probabilities are periodic that introduces picks in the Fourier Transform of probability map



Detecting traces of resampling

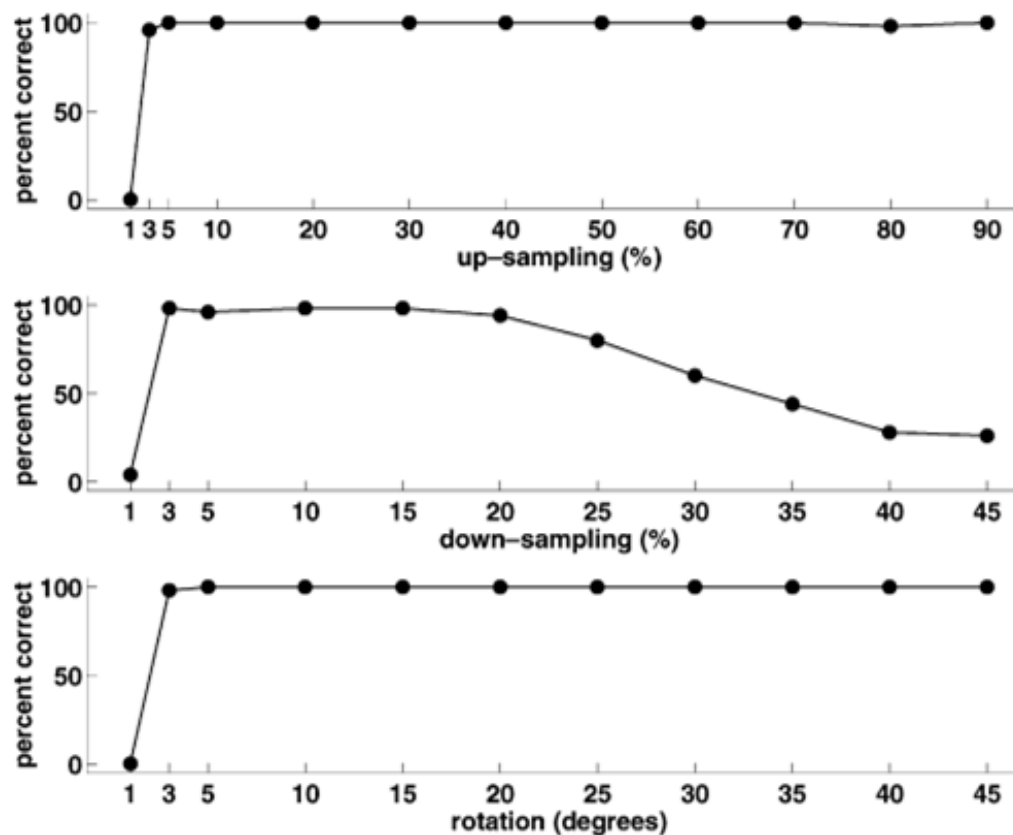
Resampling factor estimation



Detecting traces of resampling

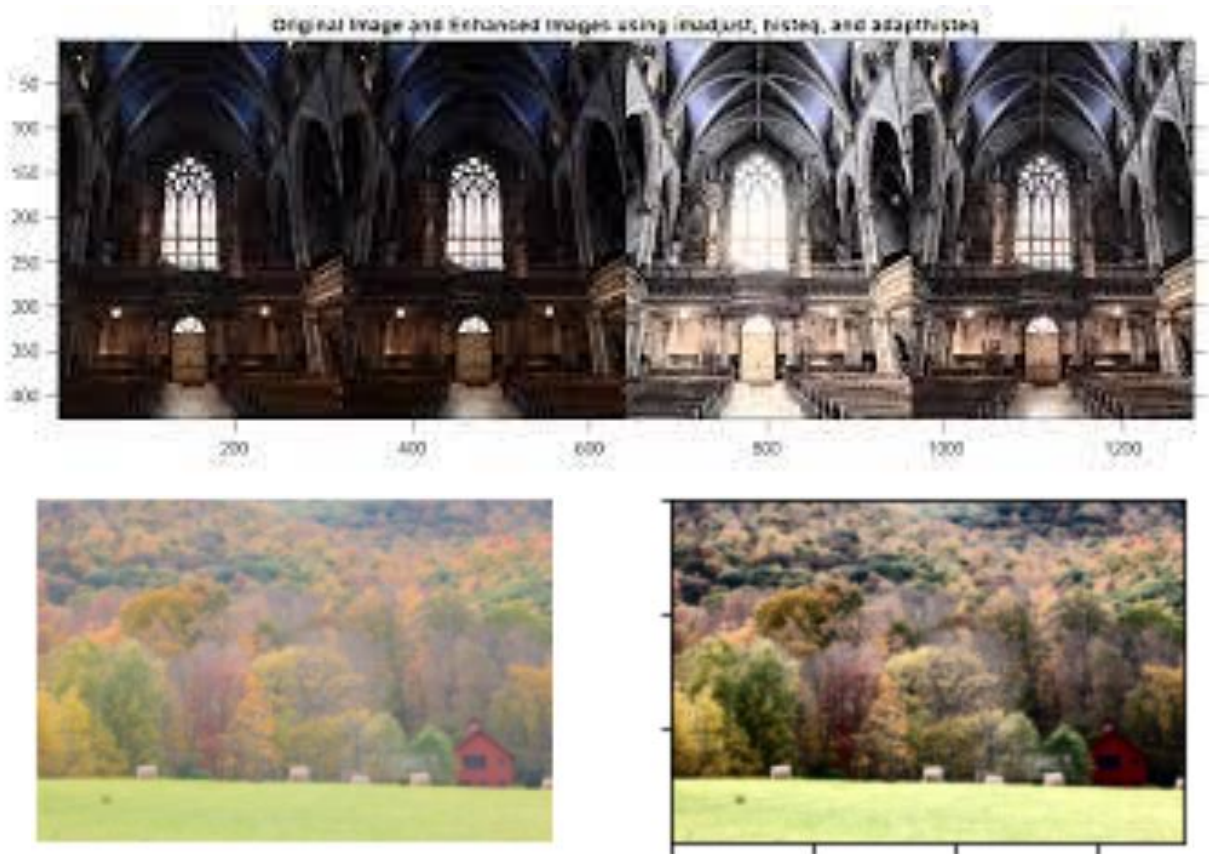
Resampling factor estimation

Average detection accuracy for 50 images



Detecting contrast enhancement

Contrast enhancement operations seek to increase the dynamic range of pixel values within images



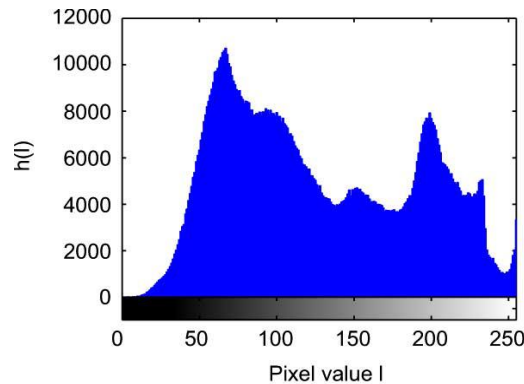
Detecting contrast enhancement

Globally applied contrast

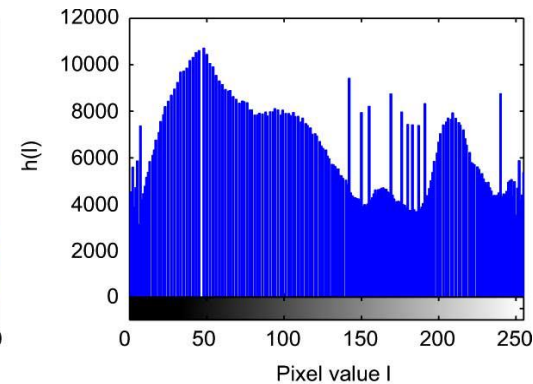
- ▶ All contrast enhancement mappings result in an increase in energy
- ▶ This energy is related to the intrinsic fingerprint
- ▶ Expected image DFTs are to be strongly low-pass signal
- ▶ Therefore, the presence of energy in the high frequency regions is indicative of contrast enhancement
- ▶ Contrast enhancement will cause isolated peaks and gaps in the histogram

Detecting contrast enhancement

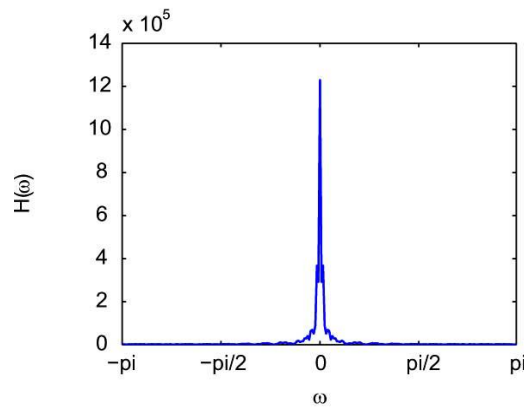
Globally applied contrast



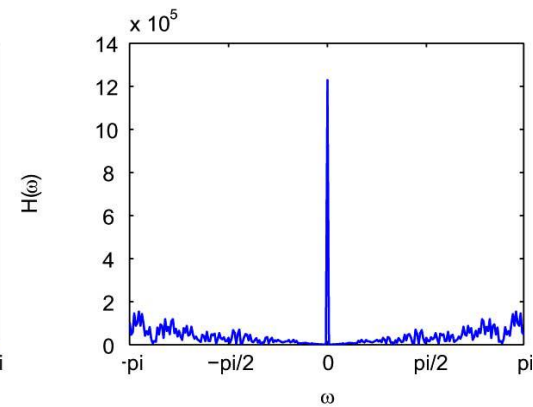
(a)



(b)



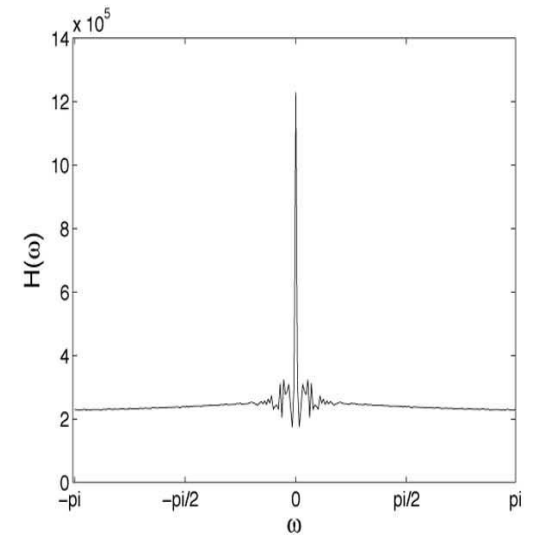
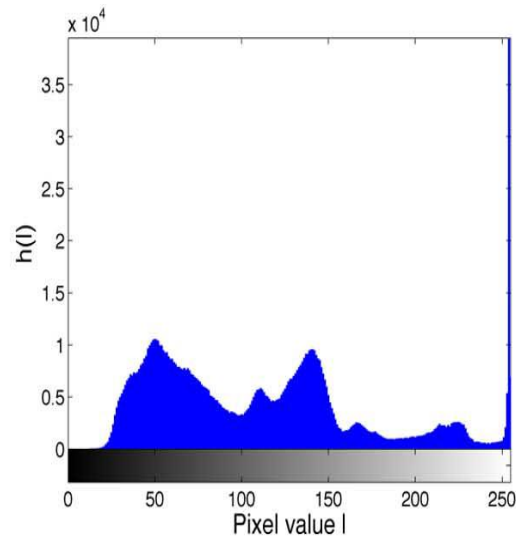
(c)



(d)

Detecting contrast enhancement

Globally applied contrast (saturation)



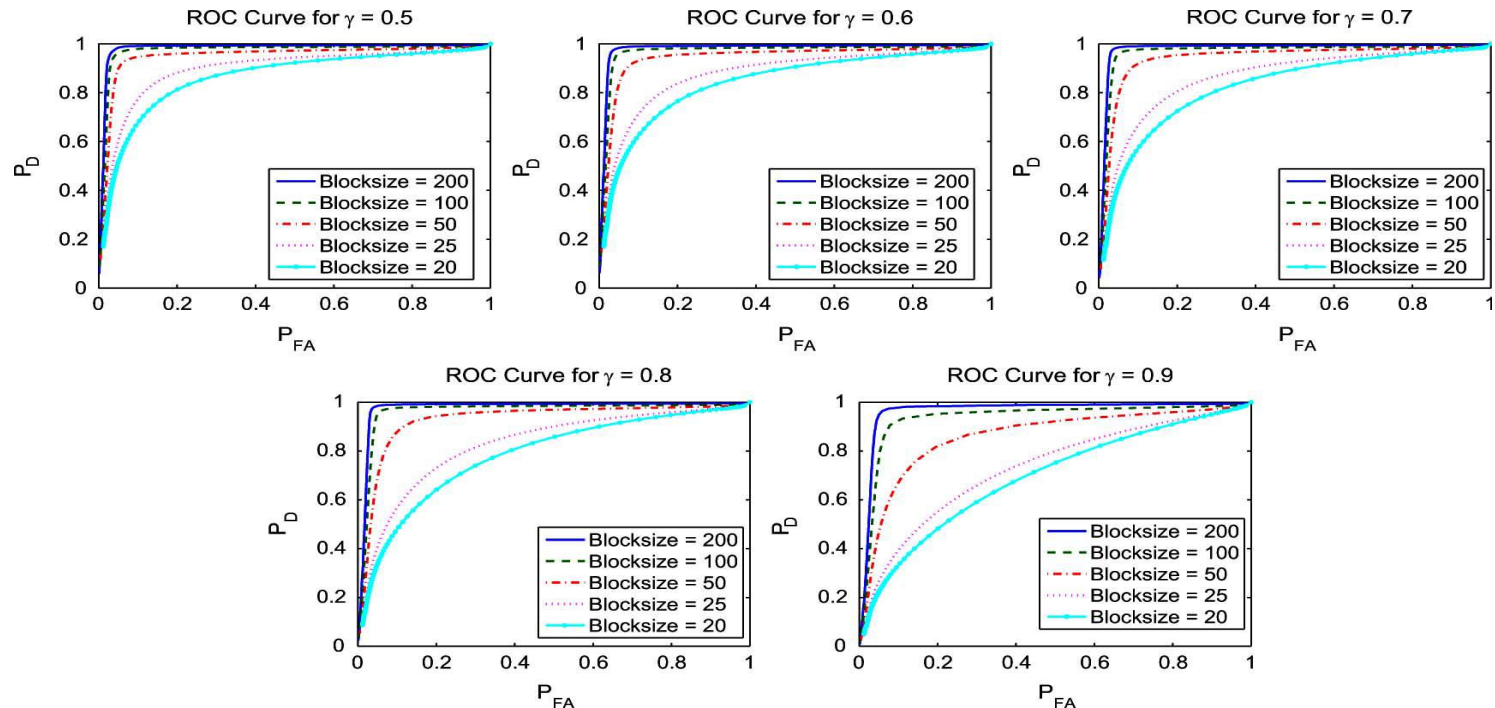
Detecting contrast enhancement

Locally applied contrast

- ▶ Defined as applying a contrast mapping to a set of contiguous pixels within an image
- ▶ Can identify cut-and-paste forgeries
- ▶ To detect it, the image is divided into smaller blocks and the global technique is applied to the blocks
- ▶ Blocks of size 200x200, 100x100, 50x50, 25x25, and 20x20

Detecting contrast enhancement

Locally applied contrast



- ▶ The contrast enhancement can be reliably detected using testing blocks sized 100x100 pixels with a P_d of at least 80% in every case at a P_{fa} of 5%
- ▶ When γ ranged from 1.0 to 2.0, the P_d was of 95% at a P_{fa} of 5%

Detecting contrast enhancement

Locally applied contrast

Local contrast for copy-and-paste detection

Images (a) and (b) are combined into the collage image (c)



(a)



(b)

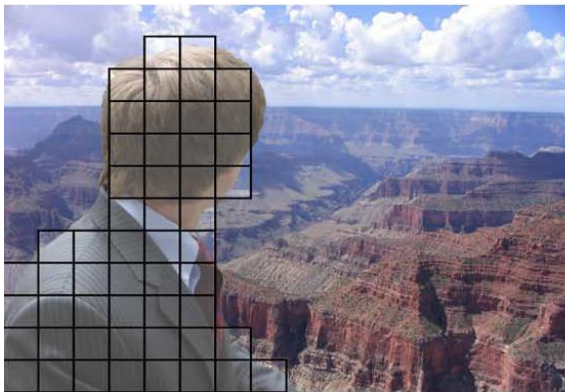


(c)

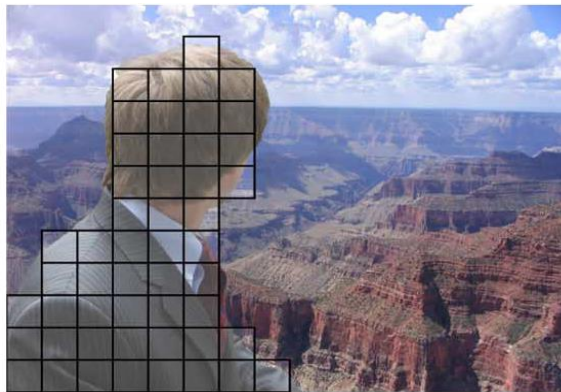
Detecting contrast enhancement

Locally applied contrast

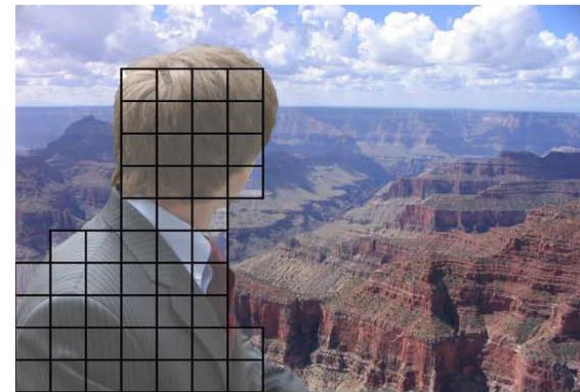
The image was divided in 100x100 pixel blocks and tested local contrast enhancement on the red (d), green (e) and blue (f) color layers



(d)



(e)

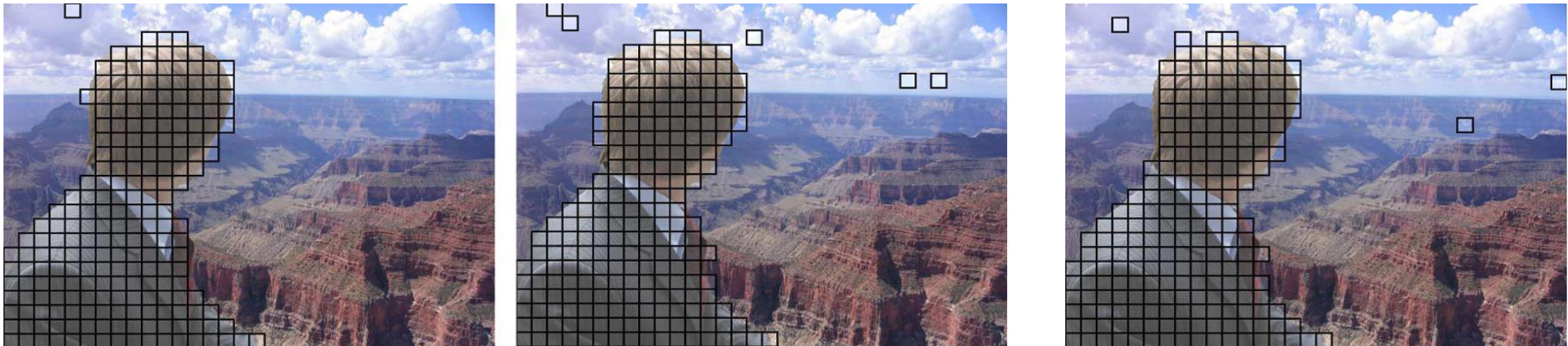


(f)

Detecting contrast enhancement

Locally applied contrast

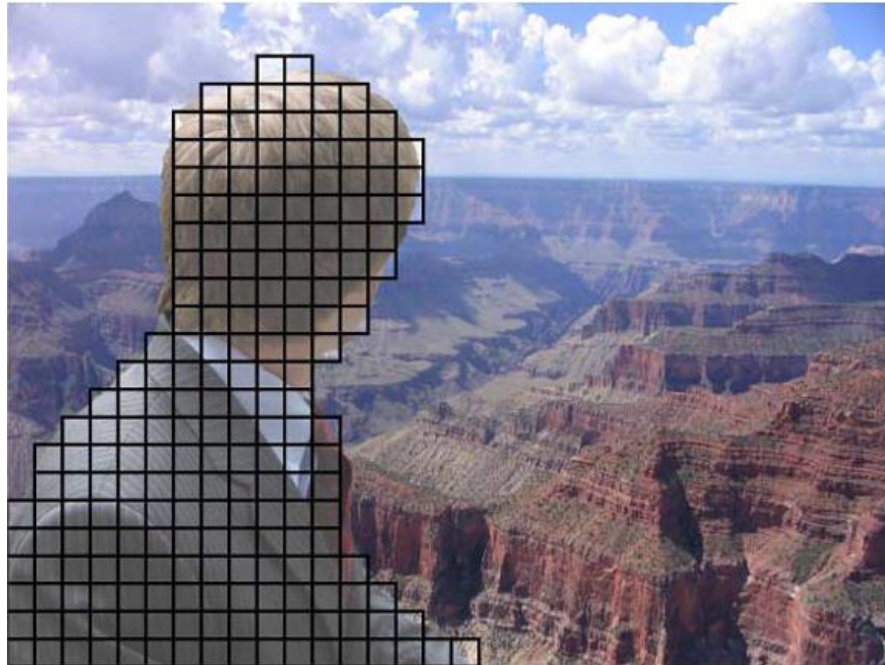
To improve localization of modified area image was divided into 50x50 pixel blocks and tested on local contrast enhancement in 3 the color layers



Detecting contrast enhancement

Locally applied contrast

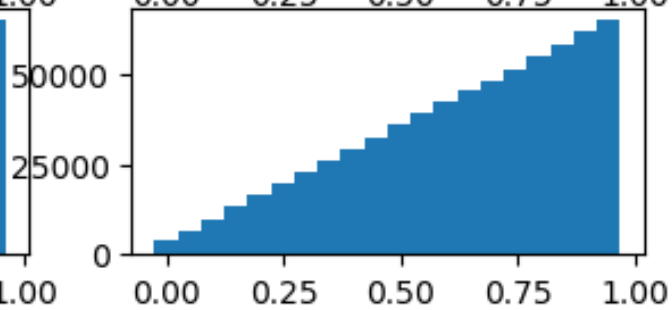
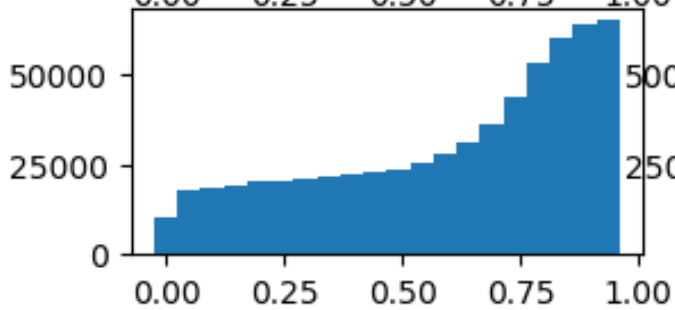
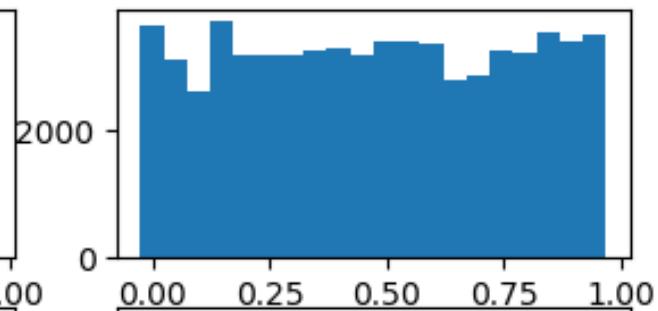
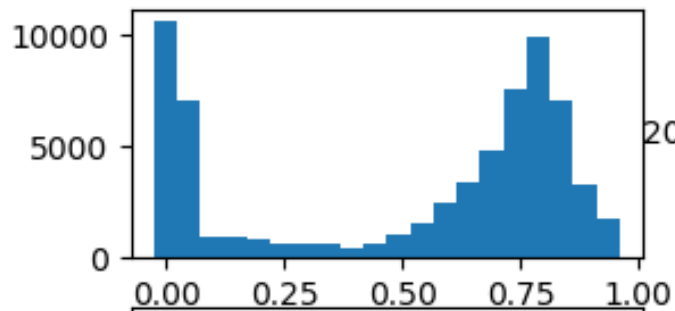
Result of detection algorithm application



Histogram equalization

- ▶ Histogram equalization effectively increases the dynamic range of an image's pixel values by subjecting them to a mapping such that the distribution of output pixel values is approximately uniform
- ▶ In order to identify it, we calculate the “uniformity” of the histogram
- ▶ For histogram equalized saturated images, the location of the impulsive component is often shifted

Histogram equalization



Tampering detection

Image manipulation (tampering): It is the application of image editing techniques to images in order to create an *illusion* or *deception* after the original photographing took place

- ▶ Tampering detection techniques can be separated in roughly 5 categories:
 - Pixel-based techniques
 - Format-based techniques
 - Camera-based techniques
 - Physics based techniques
 - Geometry-based techniques

Tampering detection

Pixel-based techniques

- Pixels are the building blocks of images
- Image manipulation disrupts statistical properties of the pixels
- Directly or indirectly analyze shows that pixel-level correlations arise in case of specific form of tampering
- Cloning: cloned regions can be of any shape and location
- Resampling: introduces specific periodic correlations between neighboring pixels
- Splicing: disrupts higher-order Fourier Statistics.
- Statistical: images contain specific statistical properties

Tampering detection

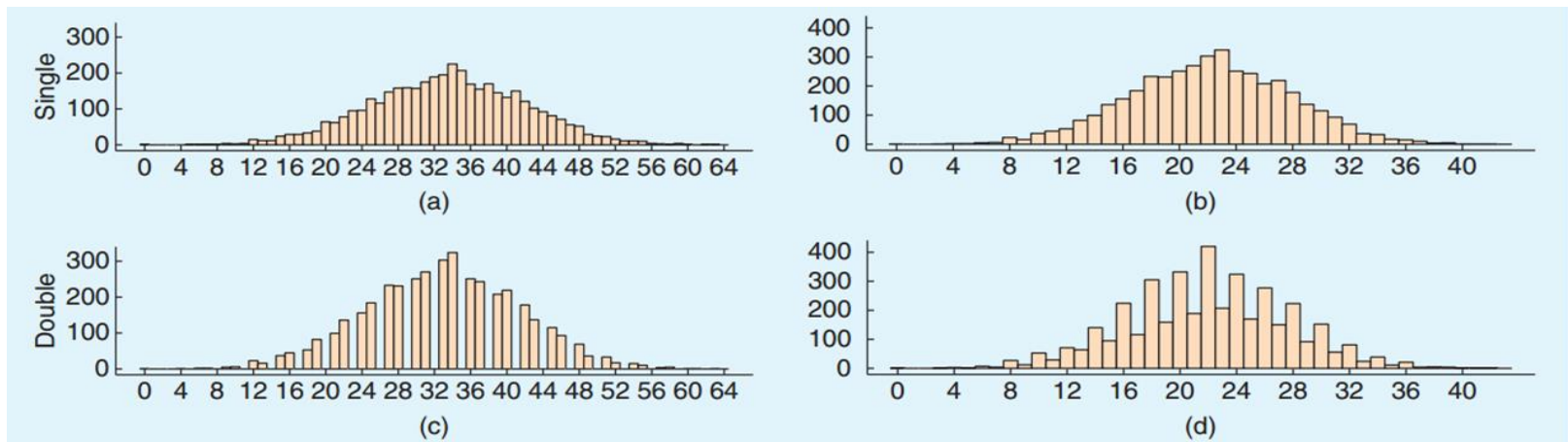
Format-based techniques

- Lossy compression introduces artifacts
- Rely on image compression specificities to detect forgery
- JPEG is the most common format

Tampering detection

Format-based techniques (examples)

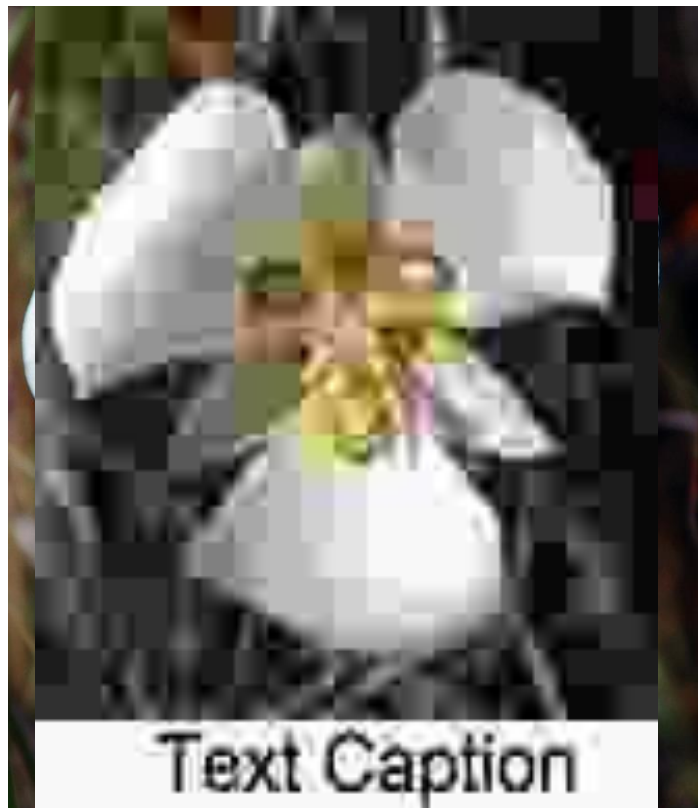
- Double JPEG compression detection



Tampering detection

Format-based techniques (examples)

- JPEG Blocking artifacts



Tampering detection

Format-based techniques (examples)

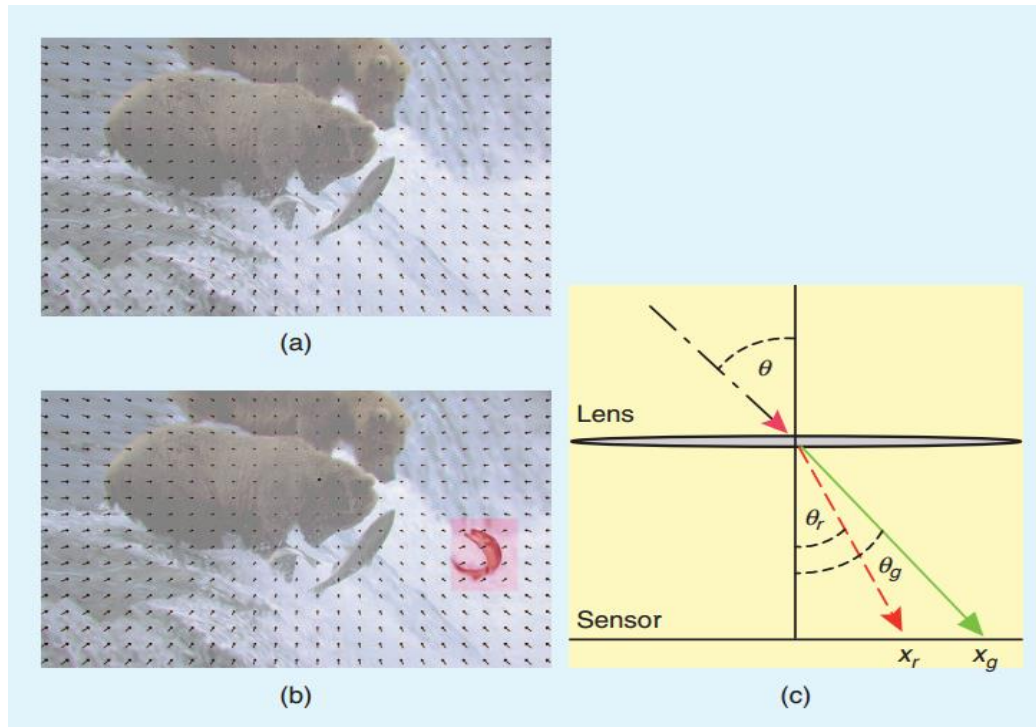
- JPEG Ghost detection: based on analyses of double JPEG compression artifacts detects lower-quality image patches spliced into higher quality images



Tampering detection

Camera-based techniques (examples)

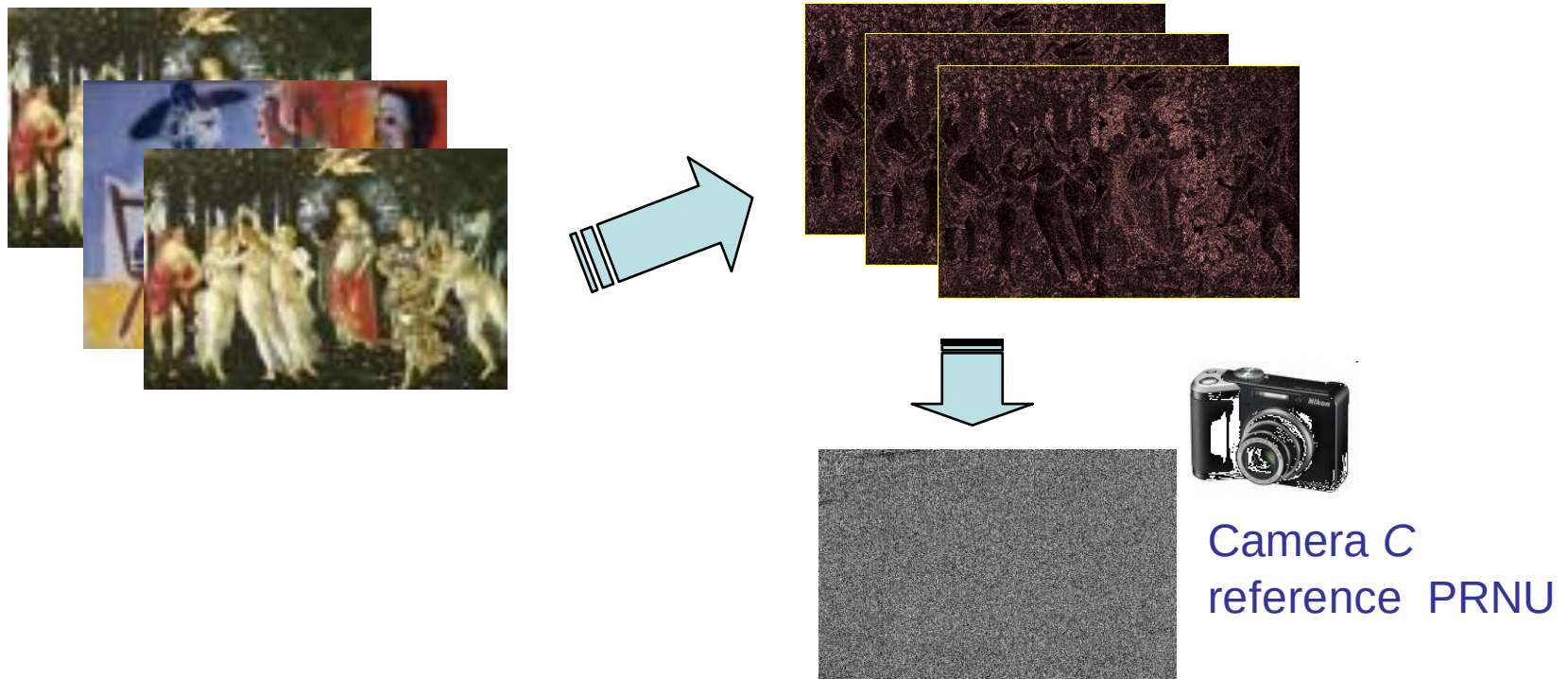
- Chromatic aberrations: variations in chromatic aberration patterns across an image may be used as evidence of image tampering



Tampering detection

Camera-based techniques (examples)

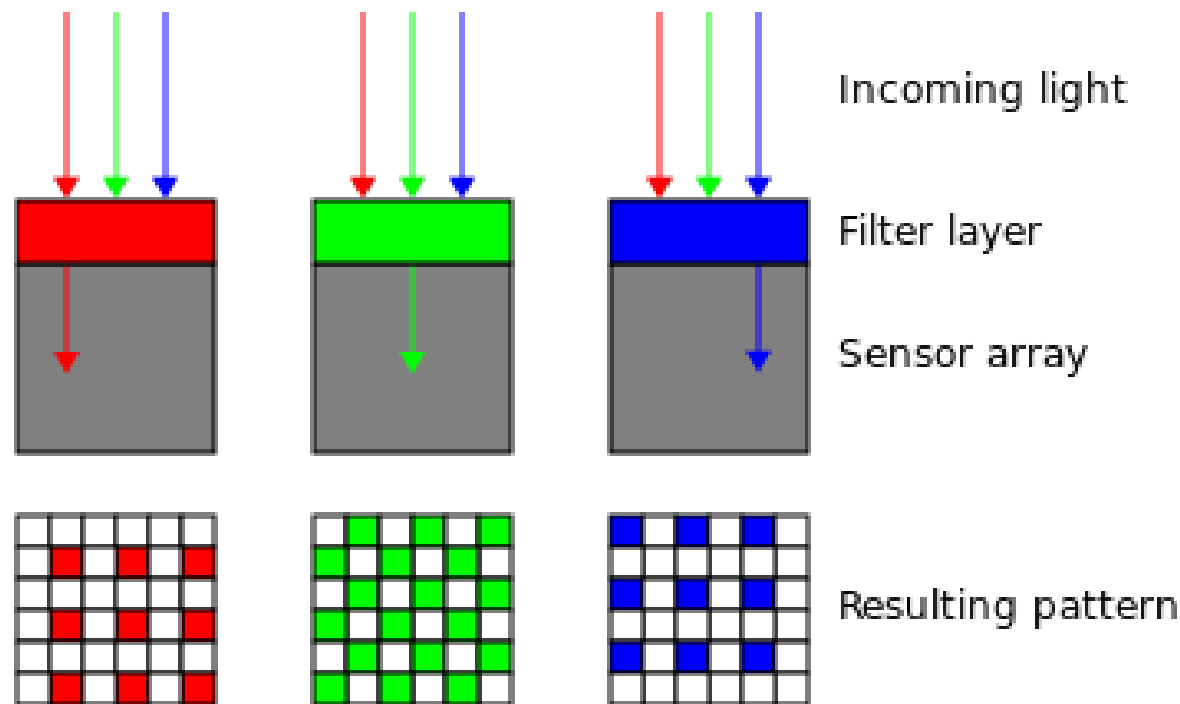
- Sensor noise: distortions of sensor noise pattern (e.g., PRNU)



Tampering detection

Camera-based techniques (examples)

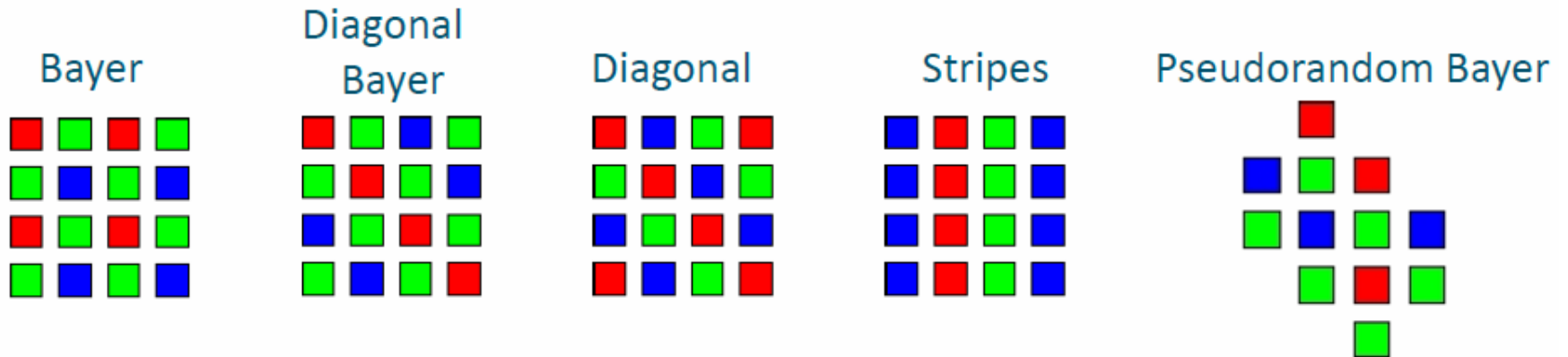
- Color filter arrays: Color calculation from neighbor pixels introduces recognizable correlation patterns between pixels in an image



Tampering detection

Camera-based techniques (examples)

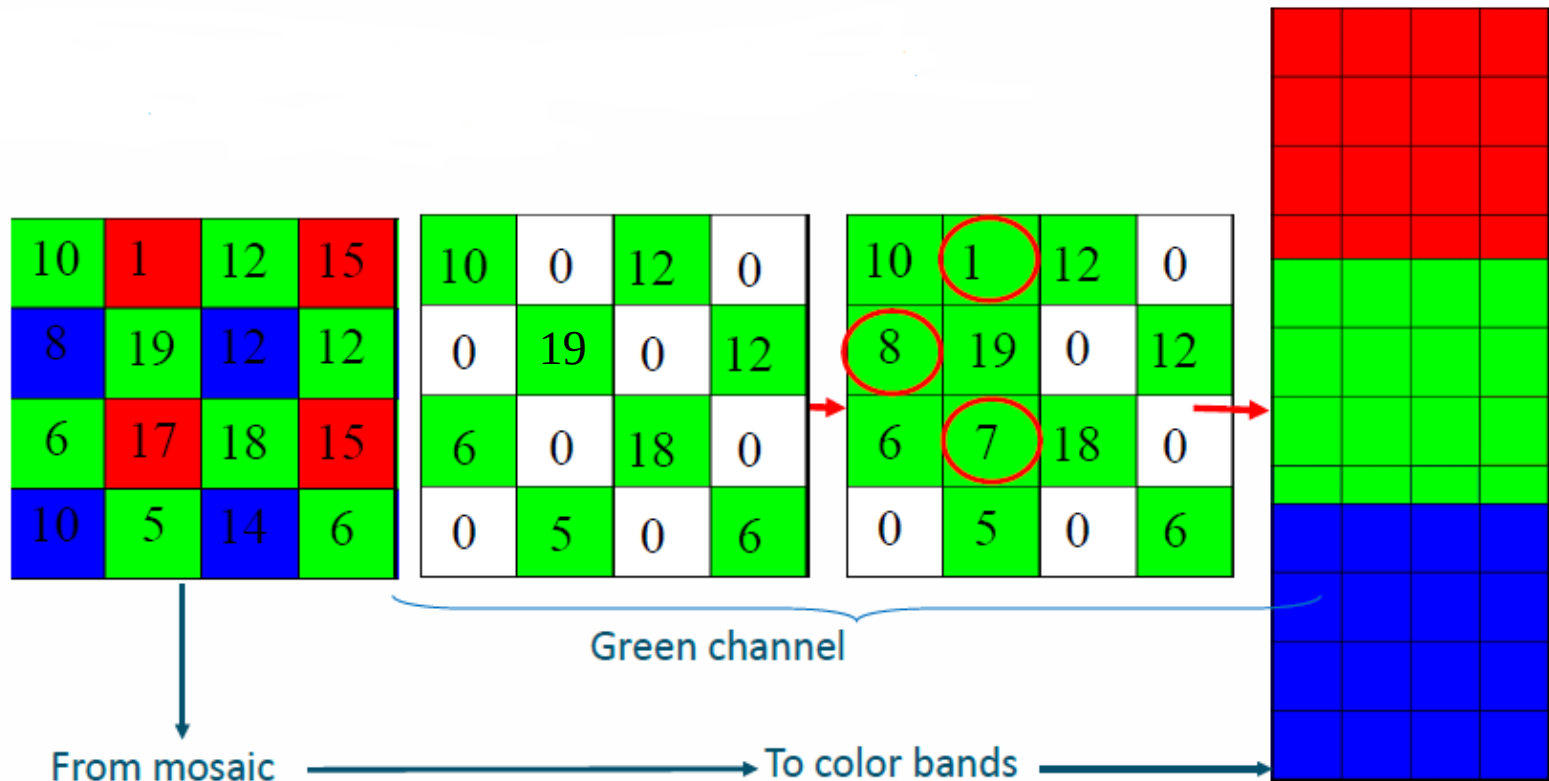
- Color filter arrays: different cameras may have different patterns



Tampering detection

Camera-based techniques (examples)

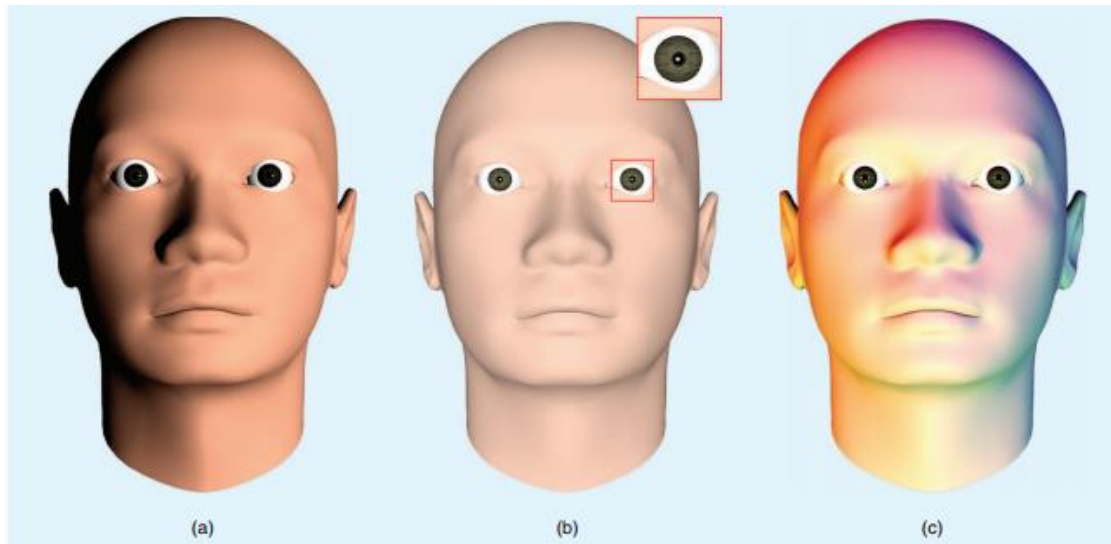
- Color filter arrays: different cameras interpolate using different, often, proprietary, filters



Tampering detection

Physics-based techniques (examples)

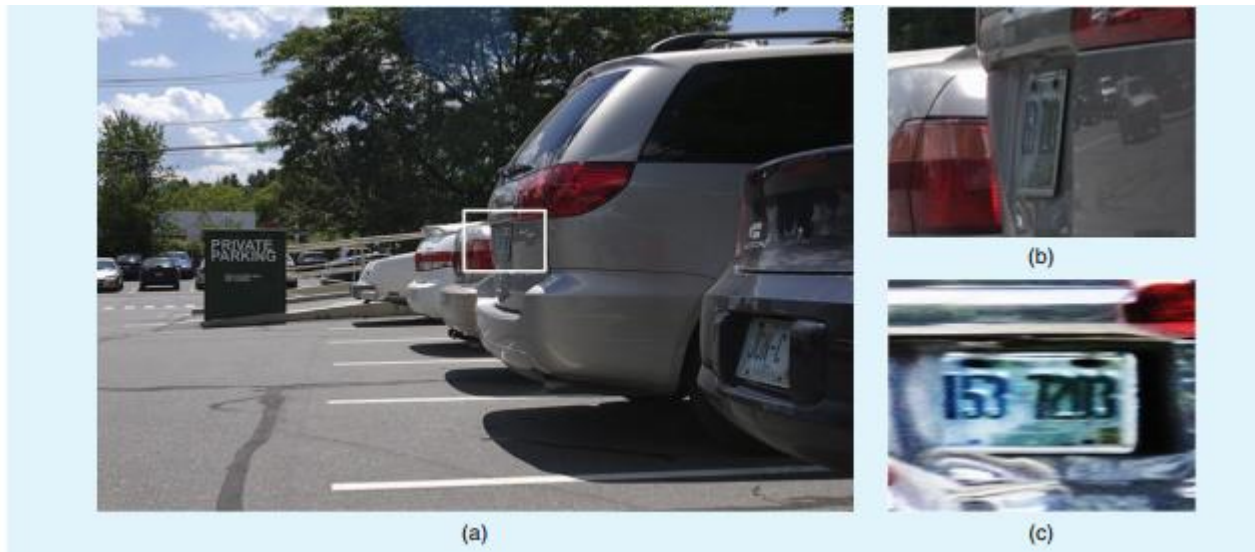
- 2-D Lighting: considers only the two-dimensional (2-D) surface normals at the occluding object boundary
- 3-D Lighting: uses the model of the human eye to determine the required 3-D surface normals
- Light environment: uses an approximation of a Lambertian surface, simplified further to consider only the occluding boundary of an object



Tampering detection

Geometric-based techniques (examples)

- Principal point estimation: principal point is the projection of the camera center onto the image plane. When a person or object is translated in the image, the principal point is moved proportionally
- Metric measurements: tools from projective geometry that allow for the rectification of planar surfaces and, under certain conditions, the ability to make real-world measurements from a planar surface



Conclusions



Original or modified?